

Data Mining Methodologies and Systems

- ❑ Structuring Unstructured Data for Knowledge Mining: A Data-Driven Approach
- ❑ Data Augmentation
- ❑ From Correlation to Causality 
- ❑ Network as a Context
- ❑ Auto-ML: Methods and Systems

From Correlation to Causality

- ❑ **Correlation analysis**

- ❑ Study the statistical associations between different observed variables

- ❑ **Causal analysis**

- ❑ Causality relation between observed variables

- ❑ **Causality analysis tasks**

- ❑ Causality discovery
 - ❑ constraint-based algorithms, score-based algorithms and functional causal models
 - ❑ Causality inference
 - ❑ Regression adjustment, propensity score methods and covariate balancing methods

From Correlation to Causality

□ Causality analysis benefits

- Enhance the interpretability of black box deep learning models
- Help avoid unfair results
- Improve the robustness of models

□ The taxonomy of causality analysis

